Review Questions

1. Copy and complete the following sentences:

(a) When one substance dissolves in another a _____ is formed. The substance which dissolves is the _____ and substance in which it dissolves is the _____.

(b) _____ is the most common solvent. Any substance which does not _____ in a solvent is said to be _____ in that solvent.

(c) Water in the solid state is called _____.

(d) To change steam to liquid water, you have to _____ it.

(e) At room temperature, water is a _____.

(f) If the temperature of water rises, it changes to a _____.

(g) When a gas cools, it may change to a _____.

2. What is the difference between a pure substance and a mixture? Give an example of each. _____

3. What method of separation would you use for the following:

(a) Iron filings spilt on the floor and swept up with dirt.

(b) Honey from honeycomb.

(c) Ball bearings from a vat of oil in a car workshop.

(d) Grading apples by size.

(e) Water from wet clothes.

(f) Salt and pepper.

(g) Oil and water.

(h) Polystyrene beads and metal bearings.

4. Give two everyday examples for each of the following situations:

(a) Filtering a solid from a gas.

(b) Filtering a solid from a liquid.

(c) Filtering gas from the air.

5. Make a list of household utensils and appliances which use a method of separation. State the method of separation used.

6. Distillation is a method used to separate the liquid part of a mixture by evaporating the solvent, collecting the solvent vapour and then condensing it to form a liquid again.

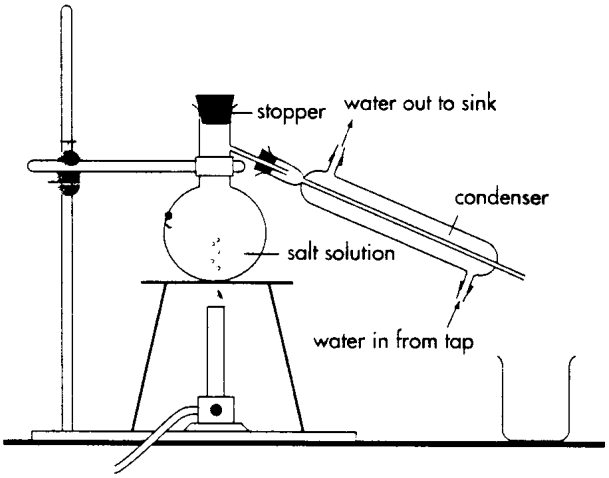


FIGURE 3.26
Experimental set-up for distillation. The solvent evaporates and is condensed in the cold delivery tube. The solute remains at the bottom of the flask.

(a) Why does steam become water as it passes through a condenser?
(b) Distillation uses two changes of state to separate and collect a liquid. What are these changes of state?

7. Complete this sentence:
A solution of sugar in water is known to be saturated when

_____.

8. Complete this crossword on this sheet.